PARTITION QUADRATIC STOCHASTIC OPERATORS CONVERGE TO THE EMPTY BODY

YITZCHAK SHMALO

Abstract. An F-quadratic stochastic operator describes a population of two sexes together with an empty body 0: parents of the same sex, or a pair containing 0, produce 0. Rozikov and Zhamilov proved that every such operator has the unique fixed point e_0 and that every trajectory converges to it exponentially fast, and Ganikhodzhaev and Rozikov asked whether the same holds when the two sexes are replaced by the blocks of an arbitrary partition. We show that it does. For a partition into q blocks the mass y outside the empty body satisfies $y' \le (1-1/q)\,y^2$, so every trajectory converges to e_0 doubly exponentially fast, and there are operators and trajectories along which this inequality is an equality at every step. The requirement that pairs containing 0 produce 0 cannot be dropped.

1. Introduction

Let $E_0 = \{0, 1, \dots, m\}$ and $S^m = \{x \in \mathbb{R}^{m+1} : x_k \ge 0,\ \sum_{k=0}^m x_k = 1\}$. A quadratic stochastic operator (QSO) is a map $V\colon S^m \to S^m$,

$$V(x)_k = \sum_{i,j=0}^{m} P_{ij,k}\, x_i x_j, \qquad k \in E_0, \tag{1.1}$$

with coefficients $P_{ij,k} = P_{ji,k} \ge 0$ and $\sum_{k=0}^m P_{ij,k} = 1$ for all i, j. The coefficient $P_{ij,k}$ is the probability that parents of types i and j have a child of type k; summing (1.1) over k shows that V maps S^m into itself. Such operators go back to Bernstein [1]; Kesten [5] studied them as models of population growth, and Ganikhodzhaev [2] the Volterra class, in which a child always has the type of one of its parents. Surveys of the theory are [4, 3]; for the role of these operators in population genetics see the books of Lyubich [6] and Rozikov [7].

Write $E = \{1, \dots, m\}$ and fix $F \subset E$, $M = E \setminus F$. Following [8, 9], the operator (1.1) is an *F-QSO* if

$$P_{ij,0} = 1 \quad \text{whenever } i, j \in F \cup \{0\} \ \text{ or } \ i, j \in M \cup \{0\}, \tag{1.2}$$

the coefficients $P_{ij,k}$ with $i \in F$, $j \in M$ being arbitrary. The type 0 is the empty body: a child is born only to a female and a male. Rozikov and Zhamilov [9] proved that every F-QSO has the unique fixed point $e_0 = (1, 0, \dots, 0)$, and that $V^n(x) \to e_0$ exponentially fast for every $x \in S^m$; this is Theorem 5 in the list of results and open problems [4]. Ganikhodzhaev and Rozikov asked the following.

Problem 1.1 ([4, Problem 10])**.** *Consider a partition $\xi = \{E_1, \dots, E_q\}$ of E. Assume $P_{ij,k} = 0$ if $i, j \in E_p$, for $p = 1, \dots, q$. Call the corresponding operator a ξ-QSO. Is an analogue of Theorem 5 true for any ξ-QSO?*

Since $\sum_k P_{ij,k} = 1$, the condition in Problem 1.1 is to be read for $k \ne 0$, so that a same-block pair produces the empty body with probability one, and the operator keeps the remaining

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requirement of (1.2), that a pair containing 0 produces 0. This gives the following definition, which for $q = 2$ is exactly (1.2) with $E_1 = F$ and $E_2 = M$.

Definition 1.2. Let $\xi = \{E_1, \dots, E_q\}$ be a partition of E into $q \geq 1$ nonempty blocks. A QSO (1.1) is a *ξ-QSO* if $P_{ij,0} = 1$ whenever $i, j \in E_p \cup \{0\}$ for some p. The coefficients $P_{ij,k}$ with $i \in E_p$, $j \in E_r$, $p \neq r$, are arbitrary.

In particular $P_{0j,0} = 1$ for every $j \in E_0$. For $x \in S^m$ put $y(x) = 1 - x_0 = \sum_{k \geq 1} x_k$, the mass outside the empty body. Since $\|x - e_0\|_1 = 2y(x)$, convergence to e_0 is the same as $y(V^n(x)) \to 0$.

Theorem 1.3. *Let V be a ξ-QSO with q blocks and put $c_q = 1 - 1/q$. Then for every $x \in S^m$*

$$y(V(x)) \leq c_q \, y(x)^2. \tag{1.3}$$

Consequently:

(i) *e_0 is the unique fixed point of V;*

(ii) *for every $x \in S^m$ and $n \geq 0$, writing $y_n = y(V^n(x))$,*

$$y_n \leq c_q^n \, y_0 \qquad \text{and, if } q \geq 2, \qquad y_n \leq c_q^{-1} \, (c_q \, y_0)^{2^n} \leq c_q^{2^n - 1};$$

if $q = 1$, then $V(x) = e_0$ for every x.

So the answer to Problem 1.1 is yes, with a rate that is doubly exponential rather than exponential; for $q = 2$ the inequality (1.3) reads $y' \leq y^2/2$. For each $q \geq 2$ the constant c_q cannot be improved, even along entire trajectories.

Proposition 1.4. *Let $q \geq 2$, let ξ consist of the singletons $E_p = \{p\}$, $p = 1, \dots, q$ (so $m = q$), and let V be the ξ-QSO with $P_{ij,k} = 1/q$ for $i \neq j$ in E and $k \in E$. If $x_1 = \cdots = x_q$, then $V(x)_1 = \cdots = V(x)_q$ and $y(V(x)) = c_q \, y(x)^2$. Hence for such x equality holds in (1.3) at every step, and $y_n = c_q^{-1}(c_q y_0)^{2^n}$ for all n.*

The requirement that pairs containing 0 produce 0 is used in the proof, and without it the conclusion fails already for $m = 1$.

Proposition 1.5. *Let $m = 1$, $q = 1$, $P_{00,0} = P_{11,0} = 1$ and $P_{01,1} = P_{10,1} = 1$, so that same-block pairs produce 0 but 0 and 1 produce 1. Then $V(x_0, x_1) = (x_0^2 + x_1^2, \, 2x_0 x_1)$ has the two fixed points e_0 and $(1/2, 1/2)$, and $V^n(x) \to (1/2, 1/2)$ for every x with $0 < x_1 < 1$.*

Without the requirement $P_{00,0} = 1$ even e_0 need not be fixed, since $V(e_0) = (P_{00,k})_{k \in E_0}$.

2. Proofs

Proof of Theorem 1.3. Fix $x \in S^m$, write $y = y(x)$ and let $s_p = \sum_{i \in E_p} x_i$ be the mass of block p, so that $s_1 + \cdots + s_q = y$. All terms of (1.1) are nonnegative, and $P_{ij,0} = 1$ for the pairs $(0, 0)$, $(0, j)$ and $(j, 0)$ with $j \in E$, and (i, j) with i, j in a common block. Dropping the cross-block pairs therefore gives

$$V(x)_0 \;\geq\; x_0^2 + 2x_0 \, y + \sum_{p=1}^{q} s_p^2.$$

By the Cauchy–Schwarz inequality $\sum_p s_p^2 \geq \frac{1}{q}\big(\sum_p s_p\big)^2 = y^2/q$. Since $x_0 = 1 - y$,

$$y(V(x)) = 1 - V(x)_0 \leq 1 - (1 - y)^2 - 2(1 - y)y - \frac{y^2}{q} = y^2 - \frac{y^2}{q} = c_q \, y^2,$$

which is (1.3).

(i) $V(e_0)_0 = P_{00,0} = 1$, so e_0 is fixed. If $V(x) = x$, then $y \leq c_q y^2 \leq c_q y$ because $0 \leq y \leq 1$, and $c_q < 1$ forces $y = 0$, that is, $x = e_0$.

(ii) If $q = 1$ then $c_q = 0$ and (1.3) gives $y(V(x)) = 0$. Let $q \geq 2$. From (1.3) and $y_n \leq 1$ we get $y_{n+1} \leq c_q y_n$, hence $y_n \leq c_q^n y_0$. Putting $z_n = c_q y_n$, (1.3) becomes $z_{n+1} \leq z_n^2$, so $z_n \leq z_0^{2^n}$ by induction, which is the second bound; the third follows from $y_0 \leq 1$. □

The proof shows when (1.3) is an equality at a point x with $y(x) > 0$: exactly when the q block masses are equal and $P_{ij,0} x_i x_j = 0$ for every cross-block pair.

Proof of Proposition 1.4. The coefficients are those of a ξ-QSO: pairs containing 0 and pairs (i, i) produce 0, and for $i \neq j$ in E the row $(P_{ij,k})_k = (0, 1/q, \ldots, 1/q)$ sums to one. Let $x_1 = \cdots = x_q = y/q$. Then

$$V(x)_0 = x_0^2 + 2x_0 y + \sum_{i=1}^{q} x_i^2 = (1-y)^2 + 2(1-y)y + \frac{y^2}{q}, \qquad V(x)_k = \frac{1}{q} \sum_{i \neq j \in E} x_i x_j = \frac{1}{q}\Big(y^2 - \frac{y^2}{q}\Big)$$

for $k = 1, \ldots, q$. The second expression does not depend on k, and the first gives $1 - V(x)_0 = y^2 - y^2/q = c_q y^2$. The property $x_1 = \cdots = x_q$ is thus preserved, equality holds at every step, and $z_n = c_q y_n$ satisfies $z_{n+1} = z_n^2$, so $y_n = c_q^{-1}(c_q y_0)^{2^n}$. □

Proof of Proposition 1.5. By (1.1), $V(x)_0 = P_{00,0} x_0^2 + P_{11,0} x_1^2 = x_0^2 + x_1^2$ and $V(x)_1 = 2P_{01,1} x_0 x_1 = 2x_0 x_1$. Both e_0 and $(1/2, 1/2)$ are fixed. With $t_n = V^n(x)_1$ the map is $t_{n+1} = 2t_n(1 - t_n)$, and $1/2 - 2t(1-t) = 2(1/2 - t)^2$. Hence $u_n = 1/2 - t_n$ satisfies $2u_{n+1} = (2u_n)^2$, so $2u_n = (2u_0)^{2^n}$. If $0 < x_1 < 1$ then $|2u_0| < 1$, and $u_n \to 0$. □

3. Computer checks

The script `code/checks.py` in [10] checks (1.3) in exact rational arithmetic on 3,200 points of 400 randomly generated ξ-QSOs with $2 \leq m \leq 7$ and $1 \leq q \leq m$, and along five steps of 40 trajectories of operators with $m \leq 4$; it checks the equality of Proposition 1.4 along four steps for $q = 2, \ldots, 6$, and the fixed point of Proposition 1.5. The file `lean/XiQSO.lean` contains Lean 4 proofs, with Mathlib, of the deduction of (1.3) from the lower bound for $V(x)_0$ (as an inequality between real numbers, with the block masses as a function on $\{1, \ldots, q\}$), of the bound $y_n \leq c^n y_0$ from $y_{n+1} \leq c\, y_n^2$, and of the conclusion $y = 0$ from $y \leq c\, y^2$ with $c < 1$. None of this is used in the proofs above.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [11] and working together inside the Brain framework [13], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [12], the Math Brain repository [11] and the Brain framework repository [13].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. On 9 September 2026 a Claude agent (Anthropic), working in Claude Code, found the inequality (1.3) and its proof, with numerical checks and a Lean formalization of the one-step bound. On 18 October 2026 Claude Opus 5.5, working in Claude Code, checked the proof line by line, added Propositions 1.4 and 1.5, and wrote the computer checks, the Lean file and the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at `https://github.com/yspennstate/partition-qso-empty-body`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [11].

References

[1] S. N. Bernstein, *Solution of a mathematical problem connected with the theory of heredity*, Ann. Math. Statistics **13** (1942), 53–61.

[2] R. N. Ganikhodzhaev, *Quadratic stochastic operators, Lyapunov functions and tournaments*, Russian Acad. Sci. Sb. Math. **76** (1993), 489–506.

[3] R. N. Ganikhodzhaev, F. M. Mukhamedov and U. A. Rozikov, *Quadratic stochastic operators and processes: results and open problems*, Infin. Dimens. Anal. Quantum Probab. Relat. Top. **14** (2011), 279–335.

[4] R. N. Ganikhodzhaev and U. A. Rozikov, *Quadratic stochastic operators: results and open problems*, arXiv:0902.4207 (2009).

[5] H. Kesten, *Quadratic transformations: a model for population growth. I*, Adv. Appl. Probab. **2** (1970), 1–82.

[6] Yu. I. Lyubich, *Mathematical Structures in Population Genetics*, Biomathematics, Springer, Berlin, 1992.

[7] U. A. Rozikov, *Population Dynamics: Algebraic and Probabilistic Approach*, World Scientific, Singapore, 2020.

[8] U. A. Rozikov and U. U. Jamilov, *On F-quadratic stochastic operators*, arXiv:math/0612225 (2006).

[9] U. A. Rozikov and U. U. Zhamilov, *F-quadratic stochastic operators*, Math. Notes **83** (2008), 554–559.

[10] Y. Shmalo, *Code for partition quadratic stochastic operators*, `https://github.com/yspennstate/partition-qso-empty-body`, 2026.

[11] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[12] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.

[13] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.

Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel

Email address: `yitzchak.shmalo@gmail.com`